Candidate's seat no.: _______________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Third Semester of B. Tech. (EC) Theory Examination Nov-2019
MA243.01 Transforms and Vector Differential Calculus

Date: 04/11/2019 (Monday) Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 70

Instructions:

i. The question paper contains two sections.
ii. Section I and II must be attempted in separate answer sheets.
iii. Figures to the right indicate marks.
iv. Make suitable assumptions and draw neat figure where it is required.
v. All notations and terminologies are standard.

Section-I

Q-1 Choose the correct answer from the given options in the followings: [07]

1) The period of $\cos pt$ is ________.
(A) 2π (B) $\frac{2\pi}{p}$ (C) $\frac{\pi}{p}$ (D) π

2) For the function $f(x)$, the one of the Dirichlet's condition is "The function $f(x)$ must be ______".
(A) single valued (B) multi valued (C) real valued (D) none of these

3) If $f(x) = x\cos x$ in $(-\pi, \pi)$ then the value of the Fourier coefficient a_n equals to ________.
(A) 0 (B) $\frac{\pi}{2}$ (C) $-\pi$ (D) 2π

4) The Laplace transform of e^{2t+3} is _________.
(A) $\frac{e^2}{s-3}$ (B) $\frac{e^3}{s-2}$ $(s > 2)$ (C) $\frac{1}{s-\log 2}$ (D) $\frac{1}{s-2}$

5) The value of $1 * 1$ is ________.
(A) 1 (B) 0 (C) t^2 (D) t

6) Which of the following function is the kernel $k(t, s)$ of the Laplace transform?
(A) $\cos st$ (B) $\sin st$ (C) e^{-st} (D) se^{t^2}

7) The inverse Laplace transform of 1 is ________.
(A) 1 (B) $\delta(t-a)$ (C) $\delta(t)$ (D) $u(t)$

Q-2 Attempt any *Four* of the followings. [16]

1) Find the Fourier series of $f(x) = x + x^2$ in the interval $(-\pi, \pi)$.

2) Obtain the half range cosine series of $f(x) = \begin{cases} 1 & , 0 \le x \le 1 \\ x & , 1 < x \le 2 \end{cases}$.

3) Find the half range sine series of $f(x) = lx - x^2$ in the interval $(0, l)$ and hence deduce that $\frac{\pi^3}{32} = 1 - \frac{1}{3^3} + \frac{1}{5^3} - \dots$.

4) Prove that $L\{\sinh at\} = \frac{a}{s^2-a^2}$ and $L\{\cosh at\} = \frac{s}{s^2-a^2}$.

5) Evaluate the Laplace transform of (i) $t\,e^{-t}\cos t$ (ii) $\frac{e^{-t}\sin t}{t}$.

6) Find $(i)\ L\{e^{-2t}\,u(t-3)\}$ $\quad (ii)\ L^{-1}\left\{\frac{se^{-2s}}{s^2+16}\right\}$.

Q-3 (a) Attempt any *Two* of the followings. **[06]**

1) Obtain the Fourier series of $f(x)=\begin{cases}-k\ ; -\pi<x<0\\ \ k\ ;\ 0<x<\pi\end{cases}$.

2) Find the Laplace transform of periodic function $f(t)=|\sin\omega t|\,; t\geq 0$.

3) Using convolution theorem, evaluate $L^{-1}\left\{\frac{a}{s^2(s^2+a^2)}\right\}$.

Q-3 (b) Attempt any *One* of the followings. **[06]**

1) Solve the differential equation $y''-3y'+2y=4t,\ y(0)=1\ and\ y'(0)=-1$ using Laplace transform.

2) Find the Fourier transform of $f(x)=\begin{cases}1-x^2\ ,\ |x|\leq 1\\ 0\ ,\quad |x|>1\end{cases}$ and hence evaluate $\int_0^\infty\left(\frac{x\cos x-\sin x}{x^3}\right)\cos\frac{x}{2}\,dx$.

Section-II

Q-4 Choose correct answer from the given options in the followings: **[07]**

1) Let the Fourier transform of a function $f(x)$ be $F(s)$. The Fourier transform of $f'(x)$ is ________.

(A) $\frac{dF(s)}{ds}$ (B) $i2\pi sF(s)$ (C) $isF(s)$ (D) $\frac{F(s)}{is}$

2) The value of $\int_0^\infty\frac{sin^2x}{x^2}dx$ is __________.

(A) $\frac{3\pi}{2}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{\pi^2}{4}$ (D) $\frac{\pi}{2}$

3) The inverse Z- transform of '1' is ___________.

(A) 1 (B) $\delta(n)$ (C) $u\{n\}$ (D) $\frac{z}{z-1}$

4) For the signal 'u_n', if $Z[u_n]=U(z)$, then $Z[u_{n-k}]$ is ________,$(k>0)$.

(A) z^k (B) z^{-k} (C) $z^kU(z)$ (D) $z^{-k}U(z)$

5) If $\bar{r}=x\hat{\imath}+y\hat{\jmath}+z\hat{k}$ then the magnitude of the vector $r^2\bar{r}$ is __________.

(A) $(x^2+y^2+z^2)^{\frac{3}{2}}$ (B) $(x^2+y^2+z^2)^2$
(C) $(x^2+y^2+z^2)^3$ (D) $x^2+y^2+z^2$

6) For which value of a constant 'k' the vector $\bar{F}=(2y-3z)\hat{\imath}-(ky+1)\hat{\jmath}+2\hat{k}$ is solenoidal?

(A) -1 (B) 1 (C) 0 (D) 2

7) Let $\hat{\imath},\hat{\jmath}$ and $\hat{k}$ be unit vectors in the X,Y and Z directions respectively. Which of the following is magnitude of $\bar{a}=\hat{\imath}-\hat{k}$?

(A) 0 (B) $\sqrt{2}$ (C) 1 (D) None of these

Q-5 Attempt any *Four* of the followings. **[16]**

1) Find the Fourier sine and cosine transform of $f(x) = e^{-ax}$.

2) Find the Z-transform of a sequence $\frac{1}{n!}$ and hence find $Z\left[\frac{1}{(n-1)!}\right], Z\left[\frac{1}{(n+1)!}\right], Z\left[\frac{1}{(n+2)!}\right]$.

3) Evaluate the inverse Z-transform of $\frac{2z^2+3z}{(z-4)(z+2)}$.

4) Find the unit tangent and unit normal vector at $t = 2$ along the curve $x(t) = t^2 - 1, y(t) = 4t - 3, z(t) = 2t^2 - 6t$.

5) Define solenoidal vector field. Find the divergence and curl of the vector $\bar{f}(x, y, z) = x^2y\,\hat{\imath} + xz\,\hat{\jmath} + 2yz\,\hat{k}$ at point $(-1, 1, 1)$. Also, find the value of $\nabla \cdot (\nabla \times \bar{f})$.

6) Show that $\bar{V} = 2xyz\,\hat{\imath} + (x^2z + 2y)\,\hat{\jmath} + (x^2y)\,\hat{k}$ is the irrotational and find a scalar potential function $\phi(x, y, z)$ such that $\bar{V} = grad\,(\phi)$.

Q-6 (a) Attempt any *Two* of the followings. **[06]**

1) Find the Z - transform of n^2.

2) Prove the convolution theorem, $F\{f(x) * g(x)\} = F\{f(x)\} \cdot F\{g(x)\}$.

3) Solve the integral equation

$$\int_0^\infty f(x)\sin sx\,dx = \begin{cases} 1; 0 \le s < 1 \\ 2; 1 \le s < 2. \\ 0; \quad s \ge 2 \end{cases}$$

Q-6 (b) Attempt any *One* of the followings. **[06]**

1) Find the directional derivative of a scalar function $\phi(x, y, z) = xy^2 + yz^3$ at the point $(1, -1, 1)$, (i) along the direction $\hat{\imath} + 2\,\hat{\jmath} + 2\,\hat{k}$ and (ii) towards the point $(2, -1, 1)$. (iii) along the direction normal to the surface $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.

2) Find the Z - transform of $\cos n\theta$ and $\sin n\theta$.

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